Functions

Chapter 4

Python for Everybody
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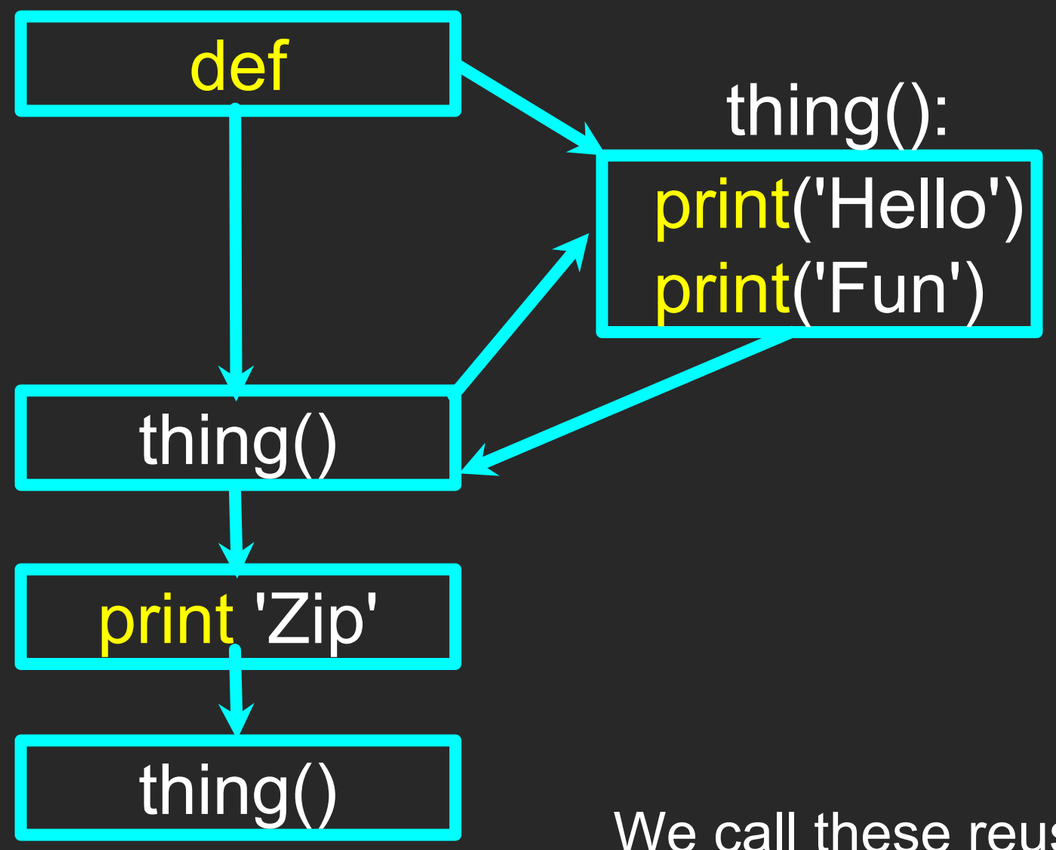
Stored (and reused) Steps



Program	Output

`def thing():`
`  print('Hello')`
`  print('Fun')`	

`thing()`	Hello Fun
`print('Zip')`	Zip
`thing()`	Hello Fun

We call these reusable pieces of code “functions”

Python Functions

- There are two kinds of functions in Python.
  - Built-in functions that are provided as part of Python - print(), input(), type(), float(), int() ...
  - Functions that we define ourselves and then use
- We treat the built-in function names as “new” reserved words (i.e., we avoid them as variable names)

Function Definition

- In Python a function is some reusable code that takes arguments(s) as input, does some computation, and then returns a result or results
- We define a function using the def reserved word
- We call/invoke the function by using the function name, parentheses, and arguments in an expression

Argument

big = max('Hello world')

Assignment

'w'

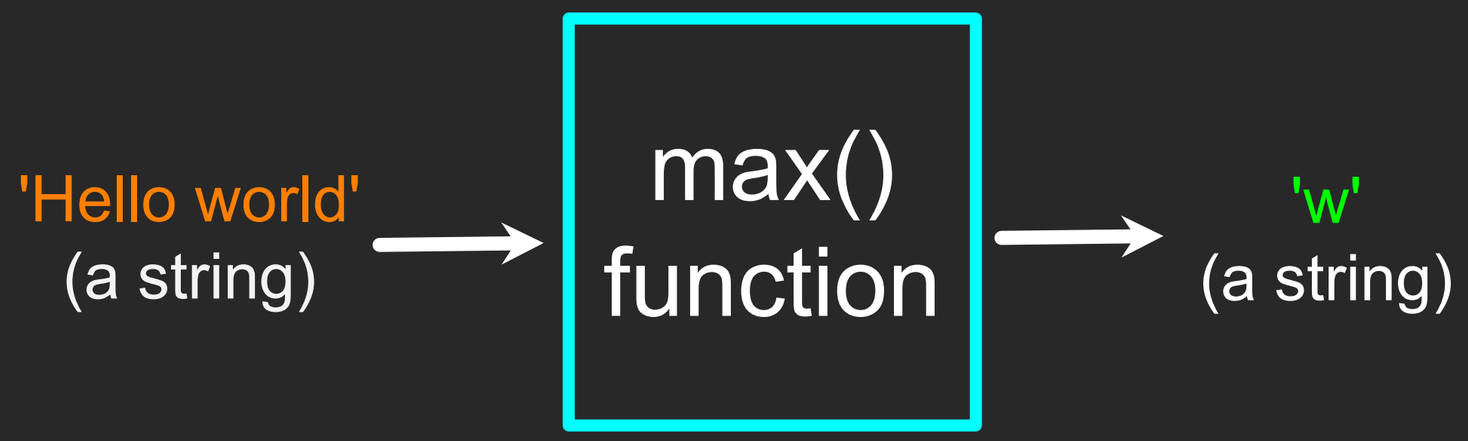
Result

```
>>> big = max('Hello world')
>>> print(big)
w
>>> tiny = min('Hello world')
>>> print(tiny)

>>>
```

Max Function

```
>>> big = max('Hello world')
>>> print(big)
w
```



A function is some stored code that we use.

A function takes some input and produces an output.

Guido wrote this code

Max Function

```
>>> big = max('Hello world')
>>> print(big)
w
```

'Hello world'
(a string)

```
def max(inp):
    blah
    blah
    for x in inp:
      blah
      blah
```

'w'
(a string)

Guido wrote this code

A function is some stored code that we use.

A function takes some input and produces an output.

Type Conversions

- When you put an integer and floating point in an expression, the integer is implicitly converted to a float
- You can control this with the built-in functions int() and float()

```
>>> print float(99) / 100
0.99
>>> i = 42
>>> type(i)
<class 'int'>
>>> f = float(i)
>>> print(f)
42.0
>>> type(f)
<class 'float'>
>>> print(1 + 2 * float(3) / 4 - 5)
-2.5
>>>
```

String Conversions

- You can also use int() and float() to convert between strings and integers
- You will get an error if the string does not contain numeric characters

```
>>> sval = '123'
>>> type(sval)
<class 'str'>
>>> print(sval + 1)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: cannot concatenate 'str'
and 'int'
>>> ival = int(sval)
>>> type(ival)
<class 'int'>
>>> print(ival + 1)
124
>>> nsv = 'hello bob'
>>> niv = int(nsv)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: invalid literal for int()
```

A Function of Our Own

Acknowledgements / Contributions

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